

Blood Glucose

CariSurg MedTech Pathways — Week 0, Day 4 Write-up

Blood glucose measures the concentration of glucose in the bloodstream, expressed in millimoles per litre (mmol/L) in most clinical settings, or milligrams per decilitre (mg/dL) in the United States. In a healthy adult, a normal fasting blood glucose falls between 4.0 and 5.9 mmol/L, and a reading taken at any time of day should generally sit below 7.8 mmol/L. At triage, blood glucose is valuable precisely because it is fast — a fingerprick test delivers a result in seconds — and because both extremes are emergencies in their own right. Hypoglycaemia (below 4.0 mmol/L) can mimic stroke, seizure, or altered consciousness, meaning a patient presenting confused or unresponsive could be misclassified entirely if glucose is not checked early. Hyperglycaemia (above 11.0 mmol/L in a non-fasting patient) may indicate uncontrolled diabetes or, in a critically ill patient, a physiological stress response to trauma or infection. Either finding changes the urgency and direction of care immediately, which is exactly what a triage metric should do.

Reference

World Health Organization. (2006). *Definition and diagnosis of diabetes mellitus and intermediate hyperglycaemia*. <https://www.who.int/publications/i/item/definition-and-diagnosis-of-diabetes-mellitus-and-intermediate-hyperglycaemia>